Koorosh Roohi

korosh.roohi9731@gmail.com | +1 (647) 594-3416 | LinkedIn | Webpage | GitHub | Hugging Face | Google Scholar

Summary

ML researcher and engineer; Ph.D. candidate at the University of Toronto and Vector Institute affiliate, with papers at NeurIPS, WACV and IEEE EMBC. Research on efficient high-resolution inference for diffusion models and on deep learning for mmWave radar and multi-modal sensing. Previously led AI product development at Masterly (VLM/LLM agents).

Experience

Vector Institute Toronto, Canada
Faculty Affiliate Researcher, Generative AI *Mar 2026 – Present*

- Research on optimizing diffusion-model inference for high-resolution output: training-free methods that let pretrained text-to-image models generate beyond their native resolution while keeping attention cost and structural artifacts in check.
- Work hands-on with text-to-image models (FLUX, Qwen-Image, Ideogram), their Diffusers implementations and weights.

Masterly Inc. Toronto, Canada
AI Manager *Nov 2023 – Sep 2026*

- Led AI projects from architecture to production on AWS (SageMaker, Bedrock, Lambda, EC2) with a 5-person team.
- Designed Masterly Earendel, an agentic VLM system that detects critical video events, alerts in real time and refines its context.
- Led FinSuite, an LLM-driven financial portal; built the MCP host and servers linking LLM/VLM agents to internal tools.

University of Toronto Toronto, Canada
Graduate Research Assistant *Apr 2023 – Present*

- Developing deep learning models that estimate infection exposure risk from indoor localization and activity recognition.
- Proposed a dynamic fusion model for mmWave radar point-cloud activity recognition, about **10%** above single-modality models.
- Built multi-modal datasets (mmWave radar, RGB-D, IMU, BLE) for activity recognition and indoor localization.

Crouse Company, *Software Engineer, Computer Vision (part-time)* Tehran, Iran · *Mar 2022 – Jun 2022*

- Built an OpenCV pipeline for defect detection in car displays, using Isolation Forest and Tesseract OCR to flag **5** defect categories.

Quiz of Kings Studio, *Client Developer, Unity (part-time)* Tehran, Iran · *Nov 2019 – Mar 2022*

- Contributed to the client of Quiz of Kings, Iran's most popular mobile game (100k+ users); led the Corona-to-Unity migration.

Education

University of Toronto Toronto, Canada
Ph.D. in Biomedical Engineering, GPA 4.0/4.0 *Sep 2023 – Aug 2027 (expected)*

- Thesis: Infection Risk Estimation via Multi-modal Sensing and Deep Learning in Healthcare

Amirkabir University of Technology (Tehran Polytechnic) Tehran, Iran
B.Sc. in Computer Engineering, GPA 3.85/4.0 *Sep 2018 – Mar 2023*

- Thesis: gait-based person re-identification (CNN, Horizontal Pyramid Matching, triplet loss), **86%** accuracy on OUMVLP-Pose.

Selected Publications

- **WACV 2027** PhotoQuilt: Training-Free Arbitrary-Resolution Photomosaics via Bootstrapped Tiled Denoising Page
- **NeurIPS 2026** SEGA: Spectral-Energy Guided Attention for Resolution Extrapolation in Diffusion Transformers Page
- **arXiv 2026** AReA: Adaptive Receptive-Field Attention for Efficient High-Resolution Diffusion Transformers Page
- **EMBC 2025** Using mmWave Radar and Deep Learning to Classify Caregiver Activities for Infection Prevention
- **Signal, Image and Video Processing 2025** FUVT: A Deep Few-shot Unsupervised Learning-based Video-to-Video Translation Scheme using Kalman Filtering and Relativistic GAN

Technical Skills

- **Machine Learning**: PyTorch, Hugging Face Diffusers, TensorFlow, scikit-learn, DiT, VLMs, LLM agents, MCP
- **Sensing & Vision**: mmWave radar point clouds, sensor fusion (RGB-D, IMU, BLE), OpenCV, YOLO, DeepSORT
- **Engineering**: Python, C#/.NET, SQL, NumPy, Pandas, AWS (SageMaker, Bedrock, Lambda, EC2), Docker, Git, Linux

Awards & Service

- **Awards**: Graduate Scholarships in Science and Technology, Mount Sinai ($15K, 2026); EPIC Doctoral Award ($10K, 2025)
- **Reviewer**: ICML 2026, NeurIPS 2026, WACV 2027, ICLR 2027; *Circuits, Systems, and Signal Processing* (13 reviews)